

AKTU UNIT 5 MASTER EXAM DECISION FLOWCHART

Step-by-Step Problem Solving Algorithms for 10-Mark AKTU Concurrency Questions (BCS501)

Step 1: Identify Concurrency Technique
Is question on Locking, Timestamps, MGL, or Oracle?

Branch A: Locking & 2PL

- Check Growing vs Shrinking Phase
- Any lock acquired after unlock? → Violates 2PL
- X-locks held till commit? → Strict 2PL
- All locks held till commit? → Rigorous 2PL
- ✓ Pre-locking all items? → Conservative 2PL

Branch B: Timestamps & Thomas Rule

- read(X): TS(T) < write_TS(X) → ABORT
- write(X): TS(T) < read_TS(X) → ABORT
- write(X) with TS(T) < write_TS(X):
 - Basic TO: ABORT T
 - Thomas Rule: IGNORE & CONTINUE (VSR)

Branch C: Multiple Granularity (MGL)

- Draw 5×5 Compatibility Matrix (IS, IX, S, SIX, X)
- Locking: Top-down from root (db)
- Unlocking: Bottom-up from leaves
- S requires IS/IX on parent
- X requires IX/SIX on parent

AKTU BCS501 Past 5-Year Question Patterns

Q1 (10 Marks): Explain Concurrent Control Mechanism & Compare 2PL vs Timestamp Ordering (2022-23)
→ Draw 2PL growing/shrinking curve + TO read/write rules. Mention Deadlocks in 2PL vs Deadlock-freedom in TO.

Q2 (10 Marks): What is Multiple Granularity Locking? Explain Compatibility Matrix & Rules (2018-19, 2021-22)
→ Draw Granularity Tree (db → f → p → r), draw 5×5 matrix, and list the 6 rules of MGL protocol.

Q3 (10 Marks): Explain Oracle Concurrency Control & Multi-Version Read Consistency (2020-21)
→ Detail SCN mechanism, Undo Segments / CR buffer reconstruction, ORA-01555 causes, and lack of lock escalation.

— DBMS Unit 5: Concurrency Control Techniques | AKTU BCS501 Semester-V —